# Felix Zhang

felixfzhang@hotmail.com | ff-zhang | felixfzhang ff-zhang.github.io

## Education

**University of Toronto** Sept. 2025 – Jan. 2027
Master of Science in Computer Science *4.0/4.0 cGPA*

**University of Toronto** Sept. 2021 – June 2025
Honours Bachelor of Science in Computer Science; Major in Mathematics *3.96/4.0 cGPA*

## Publications

**EFL: Efficient, Fault-tolerant, and Low-cost Shuffle Service with Complete DPU Offloading**
**Felix Zhang**, Joseph Zhu, Qizhen Zhang
*Under Review*

**PD3: Prefetching Data with DPUs for Disaggregated Memory** May 2026
Sidharth Sankhe, **Felix Zhang**, Umayrah Chonee, Sherman Lim, Jason Hu, Jialin Li, Qizhen Zhang
*23rd USENIX Symposium on Networked Systems Design and Implementation (NSDI '26)*

## Industry Experience

**ML Runtime Engineer**; Cerebras Systems May 2024 – Aug. 2025

- Implemented a runtime virtual memory system in **C++** which pre-emptively loads data before it is accessed, allowing off-chip memory to be used for the first time
- Added support for network storage using remote direct memory access (RDMA), providing **100** GB/s read and write speeds with **10** $\mu$s latency to multiple remote servers
- Enabled the ability log and replay the network operations, decreasing the time to recreate stalls and timeouts by over **80%**, and setup unit tests to automatically catch breakages and performance regressions in the network layer
- Improved the throughput of the network layer by **6%** when transferring data by implementing best practices for remote direct memory access (RDMA) and reducing setup overhead in **Linux**

## Research Experience

**Research Assistant**; Far Data Lab, University of Toronto Sept. 2024 – Aug. 2025

- Investigated offloading computation onto data processing unit (DPUs) to enable efficient, scalable data processing
- Implemented and parallelized the execution of *Monodepth2* in **C++**, achieving linear performance scaling with the number of threads when running on a DPU
- Built a DPU-based prefetcher *PD3* with a team of 6 which intercepts network traffic to predict and prefetch data for tiered key-value stores, eliminating the network overhead introduced when fetching entries from remote
- Designed an external service for offloading shuffle operations onto DPUs in distributed data analytics which supports both disagregated memory and storage backends

**Research Assistant**; University of Toronto March 2024 – Dec. 2024

- Worked on a team of 11 to implement a pedagogical kernel *KidneyOS* in **Rust** to be used in an introductory operating systems course with **500+** students annually
- Enabled thread creation and destruction, multi-threading, pre-emptive scheduling within the thread system
- Led a team of 3 to implement POSIX-compatible syscalls and add support for running user-space executables

**Research Assistant**; PRISM Lab, Bloorview Research Institute June 2024 – Aug. 2024
*Supervisors: Erica Floreani & Prof. Tom Chau; Funded by: FUSRP*

- Curated deep-learning models from the literature on denoising electroencephalogram (EEG) data in a team of 4 and benchmarked them on the *EEGDenoiseNet* dataset
- Investigated the applicability of end-to-end transformer models to denoise EEG signals and the impact of using signals' time-frequency representation as input on model performance

## Technical Skills

**Languages** C++, Rust, Python, C, C#, Bash, Java
**Frameworks** Catch2, PyTest, PyTorch, TensorFlow, OpenCV, scikit-learn, NumPy, Pandas, SciPy, Eigen3
**Tools** Git, Linux, Unix, tmux, CMake, Docker, Anaconda, Google Colab, QEMU, Jupyter, WSL, Slurm